

Tactile Sim2Real

Reliability of the Tactile Simulation Methods We Used

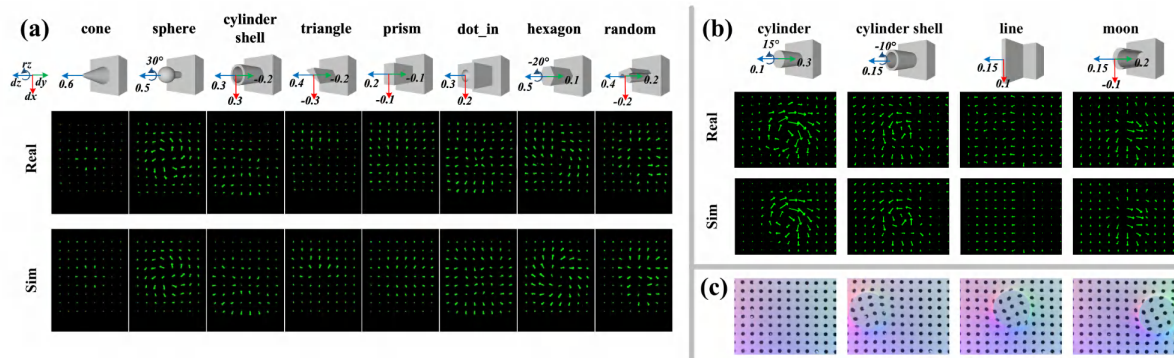
We provide the data reported for these methods in the original papers:

FOTS page 6 Fig 7 (<https://arxiv.org/pdf/2109.04027>)

"Fig. 7. (a) Marker motion field simulation results of a DIGIT sensor. We visualize the marker motions (scaled up by 5 for better visualization) on the objects from [12] under different normal, shear, and twist loads.

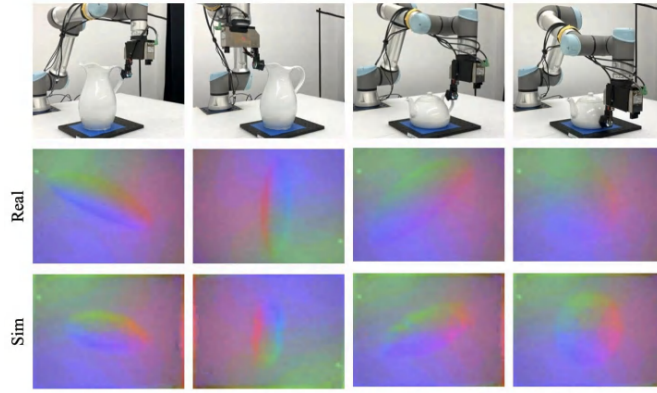
(b) Marker simulation of a GelSight sensor whose marker arrangement is different from the DIGIT sensor. (c) shows marker distortion under different loads

Fig. 7(b) shows, our marker simulation method still performs well on another type of optical tactile sensor (i.e. the GelSight) with a different marker arrangement."



Taxim page 7 Fig 11 (<https://arxiv.org/abs/2404.19217>)

"Fig. 11: Optical simulation results for objects from the Google Scan dataset [38]. We touch the objects with a GelSight mounted on a robot arm given certain contact locations. Most artifacts in the simulated images come from the coarse mesh files of the objects."



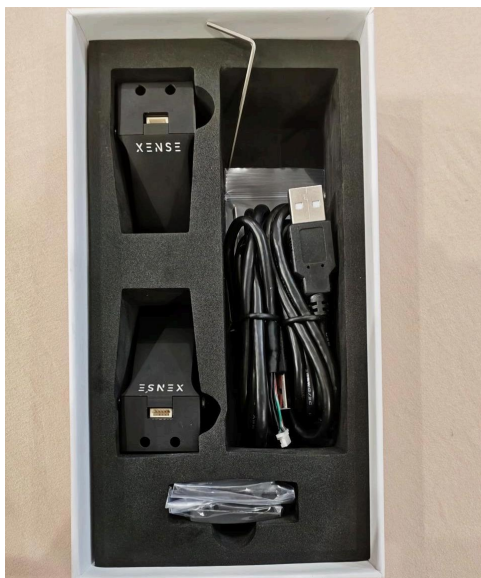
Based on the above two methods, TacEx

(<https://sites.google.com/view/tacex>) provides examples of tactile simulation, and we followed their setup. **We did not modify the GelSight sensor simulation scheme provided by TacEx in Tabero.**

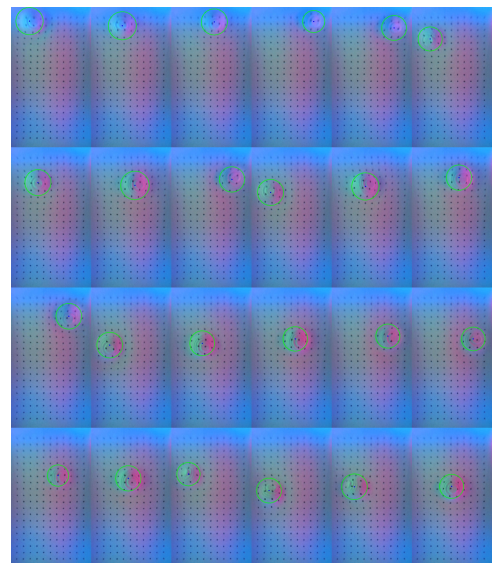
Our Verification of Tactile Sim2Real Methods

We validated these methods on our own experimental setup. The tactile sensor we used is a visuotactile sensor, model Xense.

We selected a small ball with a diameter of 6 mm and rolled it on the tactile sensor. We collected 74 images for the left finger and 89 images for the right finger, covering as many rolling positions and depths as possible.



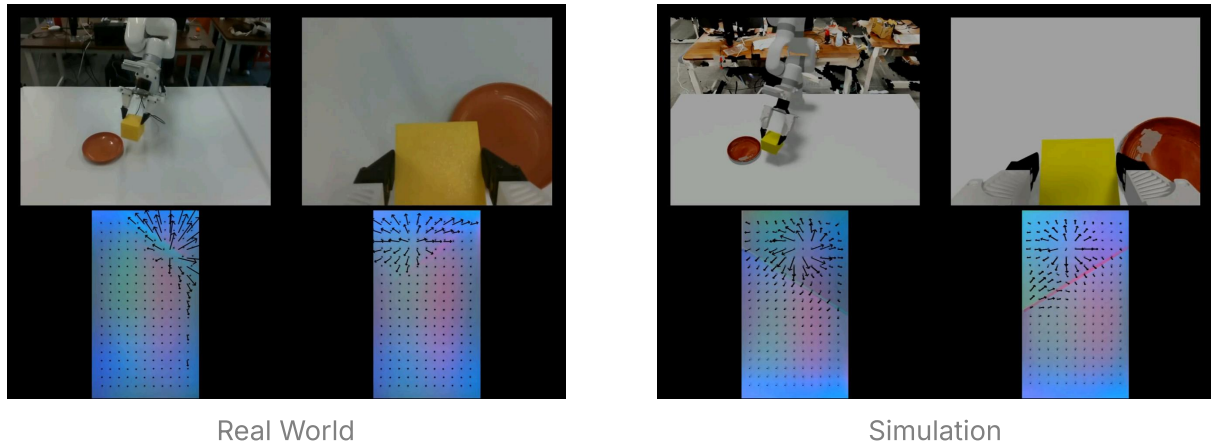
Xense Sensors



Labeled Tactile Data

We followed the settings of FOTS, Taxim and TacEx, and imported the models into Tabero for testing through calibration. We employed **the replay method proposed in our Tabero framework** to replay real-robot data in simulation and perform comparisons.

Partial test results are shown below, We uniformly scaled the arrow lengths of the marker motion field (same as FOTS) to make the results more visible.



Our reproduced results are consistent with those presented in the original papers and methods, demonstrating the reliability of the simulation. Additional results can be viewed in the videos on our website.